

Let us check the `chmod` command basic form to modify the permissions of a directory or file.

```
chmod [class][operator][permission] file name
```

```
chmod [ugoa][+ or -] [rwx] file name
```

Initially, this may appear difficult. But, it is easy. Let us start with the classes:

- **u**: This is addressed to the owner.
- **g**: This is for the group.
- **o**: This is for everyone else.
- **a**: All of the permissions above will be changed as a result of this.

Then there are the operators:

- **+**: The plus sign will add the following permissions.
- **-**: The minus sign will remove the following permissions.

The final portion is the same as when tested a file's permissions:

- **r**: Allows access to read data
- **w**: Allows access to write data
- **x**: Allows the program to run

Let us put it all together now. Assume you have a file called "sample.txt" with the following permissions: `-rw-rw-r-`

The owner and group can read and write, but the rest of the world can only read. We would want to make the following changes to the permissions: `-rwxr--`

The owner, in other words, has full permission, and the group can read. This may be accomplished in three steps. First, add the user's execution permission. `chmod u+x sample.txt`

The group's write permission will then get removed.

```
chmod g-w sample.txt
```

Finally, it will eliminate the read access for the remaining users.

```
chmod o-r sample.txt
```

You may also merge these commands into a single command, as in:

```
chmod u+x,g-w,o-r sample.txt
```

There are no spaces between the sections, which are divided by commas.

Here are a few permissions that you might find useful:

- `-rwxr-xr-x`: The owner has full permissions, while the group and other users can read and execute the file's contents.
- `-rwxr--r--`: The owner has full permissions, while the group and other users can only read the file (this is helpful if you do not mind other people looking at your files).